Ahmedabad University

ECE 501: Digital Image Processing

Week 8 Progress Report

Group 6

Vehicle Orientation Estimation Using Conventional Image Processing

Submitted to faculty: Prof. Mehul Raval

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GitHub Repository (DRASHTI-HaOBB):

https://github.com/ERYAGNIK003/DRASHTI-Dataset

Dataset Source:

DRASHTI-HaOBB – Heading-angle Oriented Bounding Box Dataset

https://zenodo.org/records/18278989

1. Work Done This Week

In Week 8, we worked on extending the vehicle axis estimation method to obtain the complete directional heading of each vehicle in the range of 0°–359°.

We adopted another silhouette-based method to estimate the front–rear direction. Otsu thresholding was used to create a binary mask and morphological closing was used to enhance the mask. The mask was rotated based on the estimated vehicle axis and the foreground pixel distribution in the left and right halves of the mask was compared. One of the two possible directions along the axis was chosen based on the calculated silhouette score.

Using this selection, the final directional heading was calculated in the range of 0°–359° for each vehicle. The estimated headings were compared with the ground-truth headings using circular angular error. Mean, median, and maximum errors were calculated to evaluate the results.

A histogram was also created to look at the distribution of the directional heading errors. Visualizations were created for all 100 vehicle observations, showing the estimated heading with a green arrow and the ground-truth heading with a red arrow. The visualizations were recorded in 5 sets of 20 vehicles.

The results, including vehicle IDs, estimated axes, estimated directional headings, ground-truth headings, and angular errors, were saved in the `Week8_Results` folder.

The Week 8 workflow was:

Loading Vehicle Crops and Ground-Truth Results → Image Preprocessing → Hough-Based Axis Estimation → Silhouette Extraction → Silhouette Asymmetry Analysis → Directional Heading Estimation → Angular Error Calculation → Visualization and Result Saving

Note: Otsu thresholding is not able to detect the actual front or rear of a vehicle. The direction estimation is based on the silhouette asymmetry and is therefore a heuristic approximation. May give false direction for symmetrical vehicles, odd shapes, overlapping objects, or poor contrast images. The estimated headings were then compared to the ground-truth heading annotations in the dataset.

2. Methodology

2.1 Loading the Previous Results and Vehicle Crops

The Week 6 vehicle crop manifest and the Week 3 results CSV were loaded using the Pandas library.

The two files were merged using the Vehicle_ID column to obtain the corresponding vehicle crop paths and ground-truth heading values.

The ground-truth heading was retained only for evaluating the estimated directional heading. It was not used during the prediction process.

The same 100 vehicle observations used in the earlier experiments were retained for Week 8. The existing reconstructed vehicle crops were reused instead of extracting a new set of images.

2.2 Vehicle Axis Estimation Using Modified Hough Transform

The Modified Hough Transform developed during Week 6 was used as the starting point for estimating the longitudinal axis of each vehicle.

The preprocessing pipeline consisted of:

1. Grayscale conversion.
2. Contrast enhancement using CLAHE.
3. Gaussian smoothing.
4. Canny edge detection.
5. Morphological closing.
6. Probabilistic Hough Line Transform.

The preprocessing settings included a CLAHE clip limit of 2.0, an 8 × 8 tile grid, a 5 × 5 Gaussian kernel, Canny thresholds of 50 and 150, and a 3 × 3 morphological closing kernel.

The Probabilistic Hough Transform was applied using the following parameters:

Parameter	Value
Hough threshold	30
Minimum line length	15 pixels
Maximum line gap	5 pixels
Minimum relative line length	0.05 × crop diagonal
Orientation bin width	20°

The orientations and lengths of the detected line segments were calculated. Line segments shorter than 0.05 times the crop diagonal were removed.

The remaining line segments were grouped into 20° orientation bins using their lengths as weights. The bin with the greatest combined line length was selected as the dominant orientation group.

A length-weighted axial mean was then calculated using the selected line segments. The resulting axis angle was represented in the range of 0°–180°.

The estimated axis was undirected, meaning that it represented the vehicle's longitudinal orientation but did not identify its front and rear.

2.3 Directional Heading Estimation Using Silhouette Asymmetry

An additional image-processing stage was implemented to address the 180° ambiguity of the estimated vehicle axis.

First, the original vehicle crop was converted into grayscale. Otsu thresholding was applied to obtain a binary image of the vehicle and its background.

Both the original binary mask and its inverse were considered. Morphological closing using a 5 × 5 kernel was applied to the candidate masks, and a suitable foreground mask was selected based on its foreground pixel ratio.

The selected mask was then rotated using the estimated vehicle axis to make the vehicle's longitudinal direction approximately horizontal.

After rotation, the image was divided into left and right halves. For each half, the number of foreground pixels in every row was calculated.

The silhouette score for each side was calculated by squaring the foreground pixel count in each row and summing the values.

The two scores were compared to select the estimated front-facing direction:

- If the right-side score was greater than or equal to the left-side score, the estimated axis angle was retained.
- Otherwise, 180° was added to the estimated axis angle.

The final directional heading was therefore obtained using one of the following expressions:

$$H = \theta \quad (if\ right\ score \geq left\ score)$$

$$H = (\theta + 180^{\circ}) \, mod \, 360^{\circ} \quad (otherwise)$$

Here, θ represents the estimated undirected vehicle axis.

This method was implemented as an experimental silhouette-based heuristic. It attempts to distinguish the front and rear using the distribution of the vehicle silhouette, but it does not guarantee correct front–rear identification for every vehicle.

2.4 Directional Heading Error Calculation

The estimated directional headings were compared with the corresponding ground-truth headings obtained from the dataset.

Since directional headings are circular measurements, the absolute difference alone cannot be used in every case. For example, the difference between 359° and 1° should be considered 2° rather than 358°.

Therefore, circular angular error was calculated as:

$$E = \min(|H_{est} - H_{GT}|, \, 360^{\circ} - |H_{est} - H_{GT}|)$$

where, H_{est} represents the estimated directional heading and H_{GT} represents the ground-truth heading.

The following performance measures were calculated:

- Number of successfully processed vehicles.
- Mean Absolute Error (MAE).
- Median angular error.
- Maximum angular error.

These measures were used to evaluate the performance of the complete directional heading estimation method.

2.5 Visual Comparison of Estimated and Ground-Truth Headings

Visualizations were generated to compare the estimated directional heading with the ground-truth heading for each vehicle.

Two arrows were drawn on the original vehicle crop:

- Green arrow: Estimated directional heading.
- Red arrow: Ground-truth directional heading.

The vehicle ID, estimated heading, ground-truth heading, and angular error were displayed along with each crop.

All 100 vehicle visualizations were saved in batches of 20 images to make the output easier to inspect and avoid generating one excessively large image file.

2.6 Saving the Results

The estimated vehicle axis, estimated directional heading, ground-truth heading, and corresponding angular error were saved in a CSV file.

The generated error histogram and vehicle visualizations were also saved in the Week8_Results folder.

The saved outputs included:

- Complete 100-vehicle directional heading results CSV.
- Directional heading error histogram.
- Directional heading visualization batch 1.
- Directional heading visualization batch 2.
- Directional heading visualization batch 3.
- Directional heading visualization batch 4.
- Directional heading visualization batch 5.

These outputs were used for the numerical and visual analysis of the Week 8 experiment.

3. Results and Observations

3.1 Loading and Processing of Vehicle Observations

The Week 8 workflow used the existing 100 vehicle observations from the previous experiments.

The vehicle crop manifest was merged with the Week 3 ground-truth results, and the corresponding vehicle crops were loaded for axis and directional heading estimation.

All 100 vehicle observations were successfully processed, and directional heading estimates were obtained for all 100 vehicles.

The same observations were retained throughout the experiment, allowing the results to be compared with the earlier orientation-estimation work.

3.2 Overall Directional Heading Estimation Results

The estimated directional headings were compared with the corresponding ground-truth headings using circular angular error.

The overall results are presented below.

Performance Measure	Week 8 Result
Total vehicle observations	100
Successfully processed vehicles	100
Unavailable predictions	0
Mean Absolute Error	82.02°
Median Angular Error	13.35°
Maximum Angular Error	179.91°
Standard Deviation	82.83°

Table 1. Overall performance of directional heading estimation on the 100-vehicle sample.

The results show that the method produced directional heading estimates for all 100 vehicle observations. However, the mean angular error was 82.02°, which indicates that the directional estimation stage did not consistently identify the correct front–rear direction.

The median angular error was 13.35°, while the maximum error was approximately 179.91°. This large maximum error indicates that some estimated headings were almost opposite to the corresponding ground-truth directions.

The difference between the mean and median errors also indicates that the error distribution was not uniform. Several vehicles produced relatively small errors, whereas a group of vehicles produced very large errors close to 180°.

This suggests that the silhouette-asymmetry method was able to produce reasonable directional estimates for some observations but did not reliably resolve the front–rear ambiguity for all vehicles.

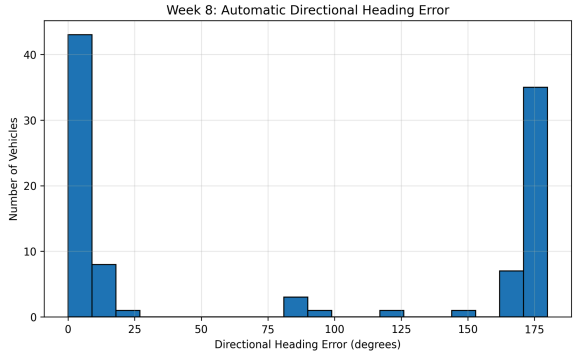


Figure 1. Histogram showing the distribution of directional heading errors across the 100 vehicle observations.

3.3 Additional Analysis of Undirected Axis Estimation

To determine if the large directional errors were due to inaccurate axis estimation or incorrect front/rear selection, another axis-only comparison was conducted with the saved Week 8 results.

The ground truth directional heading was converted to an undirected axis for this analysis by taking the value modulo 180°. This ground-truth axis was then compared to the estimated axis, taking the shortest angular difference within a 180° period.

The results were:

Performance Measure	Axis Estimation Result
Mean Axis Error	9.09°
Median Axis Error	4.46°
Maximum Axis Error	88.00°
Vehicles with axis error ≤ 30°	94

Table 2. Additional evaluation of the undirected vehicle axis using the saved Week 8 results.

The axis-only results show that the estimated longitudinal axis was considerably more accurate than the complete directional heading. Most of the vehicle observations had an axis error of 30° or less.

This comparison suggests that the main difficulty in Week 8 was not only estimating the longitudinal axis, but also selecting the correct direction along that axis.

The directional estimation stage introduced an additional source of error because it relied on silhouette asymmetry to distinguish the front and rear of the vehicle.

Note: The axis-only error analysis was calculated separately from the saved Week 8 CSV for further evaluation. It was not one of the original summary metrics printed by the Week 8 notebook.

3.4 Visual Analysis of Estimated and Ground-Truth Headings

The estimated and ground-truth directional headings were visualized for all 100 vehicle observations.

For each vehicle, a green arrow represented the estimated directional heading and a red arrow represented the ground-truth heading.

The visualizations were generated in five batches, with 20 vehicle observations in each batch.

These visual comparisons helped examine the difference between the predicted and actual vehicle directions. In cases with small angular errors, the two arrows were approximately aligned. In cases with large angular errors, the arrows showed a noticeable difference in direction, and in some cases they pointed in nearly opposite directions.

The visualizations also helped identify the effect of the front–rear selection stage on the final heading estimate.



Figure 2. Visual comparison of estimated and ground-truth directional headings for the first 20 vehicle observations.

The remaining vehicle visualizations were also generated and saved:


GgeL_IRkt_Hpl7_vehicle_1
GT: 360.0° | Est: 175.1°
Error: 175.1°


HLsx_KKFI_4Azz_vehicle_1
GT: 178.8° | Est: 352.2°
Error: 173.4°


I7oi_1x5W_O2OS_vehicle_1
GT: 268.0° | Est: 270.0°
Error: 2.0°


Iw5d_XCYw_iwrB_vehicle_1
GT: 214.7° | Est: 70.5°
Error: 144.2°


JZv9_oqt_YMRX_vehicle_1
GT: 180.0° | Est: 173.9°
Error: 6.1°

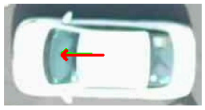

K8fi_E7BV_E33r_vehicle_1
GT: 115.2° | Est: 128.9°
Error: 13.7°


KNyT_zacV_LpYg_vehicle_1
GT: 283.2° | Est: 111.1°
Error: 172.1°

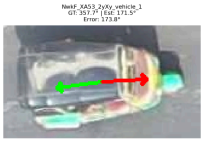

Kp67_1UBu_Rx2q_vehicle_1
GT: 330.6° | Est: 138.2°
Error: 167.6°


LGTf_z88f_Je00_vehicle_1
GT: 274.0° | Est: 271.9°
Error: 2.1°


LQKr_rvnD_oaek_vehicle_1
GT: 60.1° | Est: 55.1°
Error: 5.0°


MOL9_poaA_17Ap_vehicle_1
GT: 180.0° | Est: 181.9°
Error: 1.9°


McY4_xL3v_IDLJ_vehicle_1
GT: 358.2° | Est: 355.0°
Error: 3.2°


NwiF_XAS3_2vXy_vehicle_1
GT: 357.7° | Est: 171.5°
Error: 173.8°


ON14_gohO_rFIV_vehicle_1
GT: 183.7° | Est: 5.7°
Error: 178.0°


OTAe_cdb1_0XqF_vehicle_1
GT: 223.8° | Est: 46.0°
Error: 177.8°


OhuE_S0Tn_Qeaz_vehicle_1
GT: 173.1° | Est: 172.1°
Error: 1.0°


OyGP_UUOQ_hJtP_vehicle_1
GT: 96.0° | Est: 266.1°
Error: 179.9°


Ozpo_Q782_09Gy_vehicle_1
GT: 189.5° | Est: 12.3°
Error: 177.2°


PiNK_43IC_DcXA_vehicle_1
GT: 326.7° | Est: 134.5°
Error: 167.8°


PJJB_b1vq_Vxaw_vehicle_1
GT: 170.6° | Est: 169.8°
Error: 0.9°


PmvM_uL7n_K0k1_vehicle_1
GT: 359.1° | Est: 181.9°
Error: 177.1°


PiXzz_utpH_j6LU_vehicle_1
GT: 156.1° | Est: 70.5°
Error: 85.5°


RyLX_MLUD_U7tv_vehicle_1
GT: 187.7° | Est: 13.2°
Error: 174.5°


SRRo_F2yn_rN3o_vehicle_1
GT: 8.5° | Est: 176.6°
Error: 168.2°


TJKp_GIRz_vafn_vehicle_1
GT: 180.0° | Est: 350.1°
Error: 170.1°

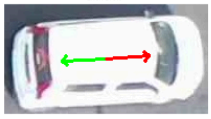

UkJR_83so_pdhs_vehicle_1
GT: 172.5° | Est: 172.5°
Error: 0.0°


Uky3_CS91_Wktl_vehicle_1
GT: 178.8° | Est: 174.7°
Error: 4.1°


WB91_Yz3z_82r4_vehicle_1
GT: 1.6° | Est: 181.4°
Error: 179.6°


Wp9e_UaVg_HSls_vehicle_1
GT: 355.5° | Est: 172.8°
Error: 177.2°


Yhs3_zETB_Lo8Y_vehicle_1
GT: 350.5° | Est: 171.2°
Error: 179.2°


ZKtx_Msfo_O7V9_vehicle_1
GT: 356.8° | Est: 177.0°
Error: 179.8°


ZP0O_Grzn_jPh0_vehicle_1
GT: 281.2° | Est: 105.8°
Error: 175.4°


a59n_AMPa_X7Go_vehicle_1
GT: 185.5° | Est: 12.5°
Error: 173.0°


bPAX_ytyN_jevf_vehicle_1
GT: 177.4° | Est: 6.3°
Error: 171.1°

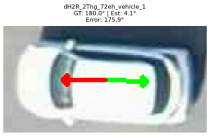

bisk_fSvT_JSaM_vehicle_1
GT: 178.8° | Est: 169.6°
Error: 9.2°


boDi_WIMN_sBBY_vehicle_1
GT: 97.0° | Est: 272.5°
Error: 175.5°


cJRl_mjsU_Z6FP_vehicle_1
GT: 0.0° | Est: 2.7°
Error: 2.7°


cLEH_vOvl_zAUO_vehicle_1
GT: 359.0° | Est: 271.0°
Error: 88.0°


dDHH_wv1m_jaew_vehicle_1
GT: 175.9° | Est: 175.2°
Error: 0.7°


dH28_2Thg_72eh_vehicle_1
GT: 180.0° | Est: 4.1°
Error: 175.9°



These images contain the remaining 80 vehicle observations and can be included in the supplementary material or appendix of the final report.

3.5 Investigation of Possible Error Sources

The Week 8 results show that the front–rear estimation stage, based on silhouette, is one of the most challenging aspects of achieving accurate directional headings.

The errors observed may be due to the following:

1. Silhouette Asymmetry Assumption

The method adopted is based on the assumption that the distribution of the foreground pixels on one side of the vehicle can be used to determine the direction of the front of the vehicle. However, the front of a vehicle does not necessarily have a larger silhouette score than its rear. This may result in the wrong direction being selected for different vehicle shapes.

2. Background Interference

Otsu thresholding is a method that divides the image based on the intensity of the pixels. It can contain shadows, road markings or foreground objects. The silhouette scores can be influenced by these extra regions.

3. Vehicle Shape and Geometry

There are various types of vehicles and each type has a different body structure. In some vehicles, the silhouette distribution of the vehicle may be almost symmetrical, and it may be difficult to determine the direction of the vehicle from the top view image.

4. Axis Alignment

The silhouette is rotated based on the estimated vehicle axis before comparing the left and right sides of the silhouette. The estimated axis and the rotation operation may be incorrect, which can impact the alignment and silhouette scores.

5. Dominant Orientation Selection

The Modified Hough method chooses the orientation bin that has the maximum total length of lines. But the best line group is not necessarily the actual longitudinal axis of the vehicle.

6. Difference Between Axis and Directional Estimation

An undirected axis is the direction of the vehicle's length but does not differentiate between the two directions along the axis. If the axis is estimated correctly, but the front/rear decision is wrong, a directional error of nearly 180° can result.

The following factors are thought to be responsible for the errors observed. A detailed analysis of the individual vehicle observations would be needed to determine the specific cause of each wrong prediction.

3.6 Observations and Limitations of the Analysis

The Week 8 experiment extended the existing orientation-estimation approach by introducing a silhouette-based method for selecting the vehicle's directional heading.

The method successfully generated predictions for all 100 vehicle observations. However, the overall mean directional heading error was relatively high at 82.02°.

The additional axis-only analysis showed a mean error of 9.09°, indicating that the undirected axis estimation was considerably better than the complete directional heading estimation on the same observations.

This difference highlights the difficulty of distinguishing the front and rear of a vehicle using only the distribution of silhouette pixels.

Another limitation is that the dominant orientation selection in the Week 8 implementation uses fixed 20° histogram bins. Angles close to 0° and 180° may fall into separate bins even though they represent nearly the same undirected axis. This may affect the axis estimation in certain cases.

The silhouette-based method was implemented as an experimental heuristic and should not be considered a fully reliable front–rear identification method.

The experiment was performed on the same 100 vehicle observations used in the previous weeks. No additional vehicle samples were introduced, and no class-wise directional heading analysis was performed during Week 8.

4. Challenges Faced

One of the main challenges during Week 8 was extending an undirected vehicle axis into a complete directional heading. The previous approaches were able to determine the longitudinal direction of the vehicle, but were unable to determine the front or rear.

Another challenge was extracting a useful vehicle silhouette from the original vehicle crops. Otsu thresholding sometimes includes background regions, which can affect the comparison between the two halves of the silhouette.

Selecting the correct direction using silhouette asymmetry was also challenging because different vehicle types have different shapes. The larger silhouette score does not necessarily indicate the front of the vehicle.

The rotation and alignment of the silhouette according to the estimated axis also affected the directional estimation stage. Any error in axis estimation or alignment could influence the comparison of the left and right sides.

Finally, evaluating the complete directional heading required circular angular error rather than the simple absolute angular difference used in the earlier 0°–90° orientation experiments.

These challenges helped us understand that estimating the vehicle's longitudinal axis and identifying its actual front-facing direction are two separate problems.

5. Current Status

The Week 8 work extended the previous orientation-estimation experiments towards the final objective of estimating the complete vehicle heading in the range of 0°–359°.

The work completed includes:

- Reusing the same 100 vehicle observations from the previous experiments.
- Loading the Week 6 vehicle crop manifest and Week 3 ground-truth heading annotations.
- Applying the Modified Hough Transform to estimate the undirected vehicle axis.
- Extracting vehicle silhouettes using Otsu thresholding and morphological closing.
- Implementing silhouette asymmetry analysis to select one of the two possible vehicle directions.
- Generating directional heading estimates in the range of 0°–359°.
- Calculating circular angular errors between estimated and ground-truth headings.
- Calculating mean, median, and maximum directional heading errors.
- Generating the directional heading error histogram.
- Generating visual comparisons for all 100 vehicle observations.
- Saving the estimated headings, ground-truth headings, angular errors, and visualizations in the Week8_Results folder.
- Performing an additional axis-only error analysis to understand the difference between axis estimation and directional heading estimation.

The experiment was successful in producing directional heading estimates for all 100 vehicle observations. The results indicated, however, that the silhouette-based front–rear selection method was not always able to select the correct direction.

The comparison between axis-only error and complete directional heading error indicated that identifying the front and rear remains the main challenge in extending the existing method to full directional heading estimation.

6. Work to Be Done Next Week

In the next stage, we plan to review the results obtained during Week 8 and investigate ways to improve the reliability of front–rear identification.

The planned work includes:

- Reviewing the vehicle observations with large directional heading errors.
- Inspecting the silhouette extraction and axis-alignment results.
- Checking the consistency of the Modified Hough implementation with the earlier Week 6 and Week 7 method.
- Investigating alternative conventional image-processing approaches for distinguishing the front and rear of vehicles.
- Comparing the results of automatic and visual front–rear identification.
- Evaluating the directional heading estimates again using circular angular error.

The next stage will focus on improving the front–rear identification process while continuing to use conventional image-processing techniques. The overall objective remains to estimate the complete vehicle heading in the range of 0°–359°.

7. References

[1] Y. Bhavsar, M. Zaveri, M. Raval, K. R. Patel, and S. B. Zaveri, "Descriptor: Drone Nadir-view Annotated Images of Vehicles Detection Dataset for India with Heading-angle Oriented Bounding Box (DRASHTI-HaOBB)," IEEE Data Descriptions, 2026. Available at: https://ieeexplore.ieee.org/document/11421776.

[2] Y. Bhavsar, M. Zaveri, M. Raval, S. Zaveri, and Ahmedabad University, "DRASHTI-HaOBB: Drone nadir-view annotated images of vehicles dataset for India - heading-angle oriented bounding box," Zenodo, 2026. Available at: https://zenodo.org/records/18278989